

LCA of impossible - Myths & Legends

SISYPHUS

THE SUM NEVER STOPS.



STEP ONE

A finite slice of eternity

FUNCTIONAL UNIT

Construction and execution of 1,000 complete punishment cycles. Each cycle pushes one reusable 5.0 t limestone boulder along a 500 m slope with 100 m vertical rise, followed by uncontrolled return to the start.

REPORTING SLICE

1,000 cycles

Eternity made calculable

REUSABLE OBJECT

5.0 t boulder

One stone for all cycles

COURSE

500 m / 100 m

Slope length / rise

THREE ACCOUNTING MODULES

01 - CONSTRUCTION

One limestone boulder represented by the DEFRA aggregate-production proxy.

02 - ASCENT

Potential-energy work plus rolling resistance, converted to an input-energy equivalent.

03 - RETURN

Gravity returns the stone. Main case assumes no input and no energy recovery.

An eternal functional unit has no finite result. The 1,000-cycle slice preserves the repeated function while keeping the calculation reportable and comparable.

STEP ONE

What Homer actually specifies

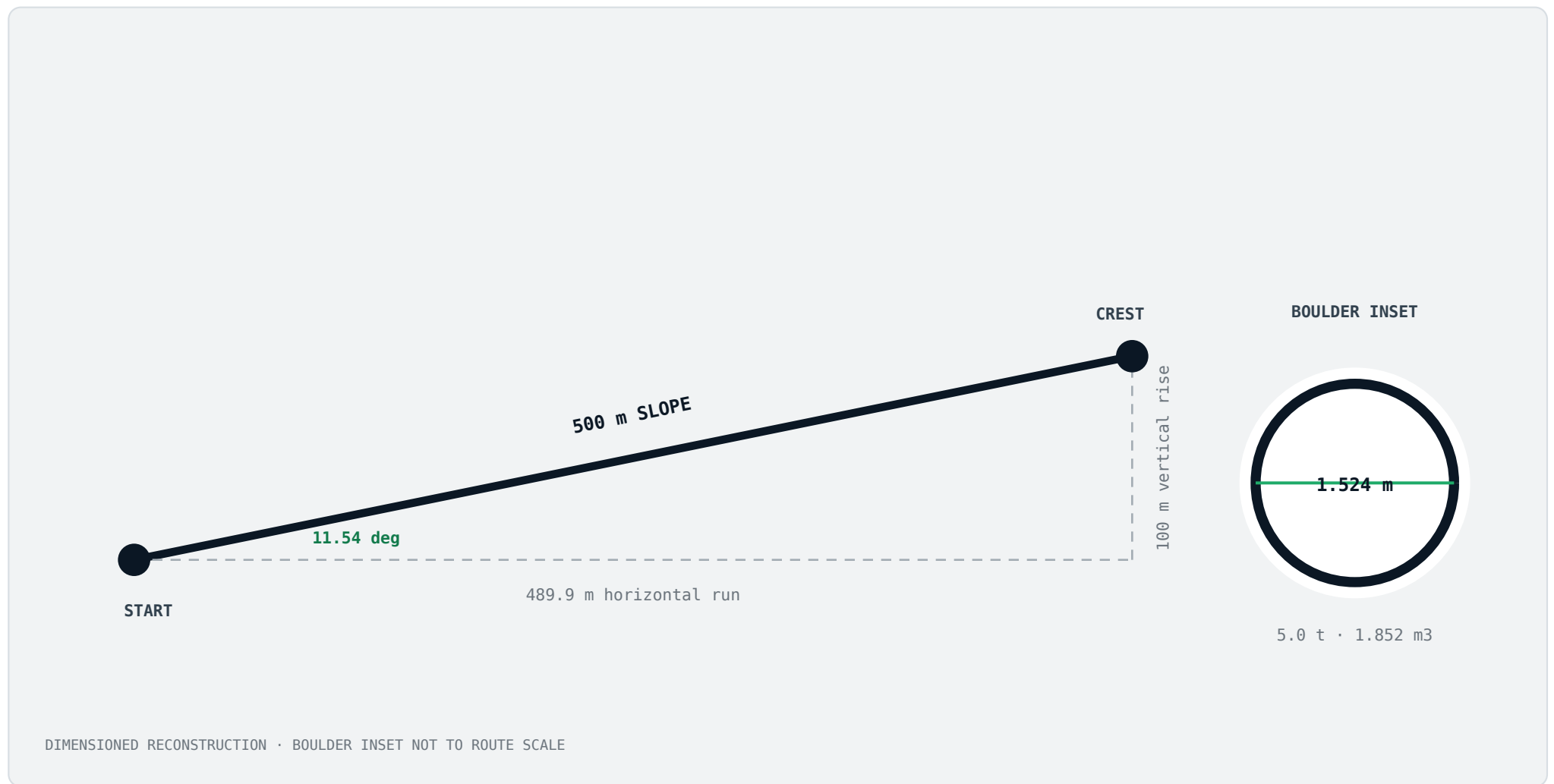
SOURCE REQUIREMENT	MODEL RESPONSE	STATUS
A monstrous stone	5.0 t limestone sphere; 1.524 m equivalent diameter	Assumed
Raised with both hands	Direct manual-contact system; no vehicle modeled	Represented
Hands and feet brace the body	Feasibility assessed through required push force	Modeled
The stone is thrust toward a hill crest	500 m incline with 100 m vertical rise	Assumed
Its weight turns it back	Uncontrolled return; no powered descent	Modeled
It rolls down to the plain	Potential energy is dissipated in the main case	Modeled
The effort restarts	1,000 ascent-return cycles as a finite reporting slice	Modeled

EVIDENCE RULE

The ancient text defines the loop. Boulder mass, geometry, resistance, efficiency and cycle count are reconstruction inputs and remain editable in the workbook.

STEP TWO

A dimensioned punishment course



SLOPE ANGLE

11.54 deg

100 m rise over 500 m

BOULDER VOLUME

1.852 m3

5,000 kg / 2,700 kg/m3

DIAMETER

1.524 m

Equivalent sphere

RECONSTRUCTION NOTES

- The route triangle is geometrically consistent: $\sin(\text{angle}) = 100 / 500$.
- The boulder inset is dimensioned separately because a 1.524 m object would be invisible at full route scale.
- Limestone density, spherical geometry and surface condition are assumptions, not archaeological claims.

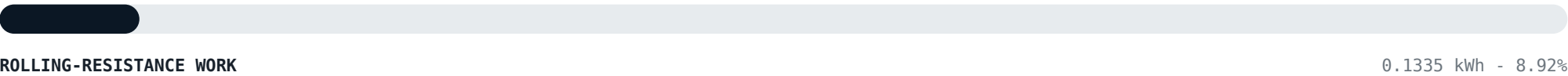
STEP TWO

One ascent: 1.496 kWh

MECHANICAL MODEL

$$E = m \times g \times h + C_{rr} \times m \times g \times \text{horizontal run}$$

TERM	CALCULATION	RESULT	SHARE
Elevation	$5,000 \times 9.81 \times 100$	1.3625 kWh	91.08%
Rolling resistance	$0.02 \times 5,000 \times 9.81 \times 489.9$	0.1335 kWh	8.92%
TOTAL MECHANICAL WORK	sum of both terms	1.4960 kWh	100.00%

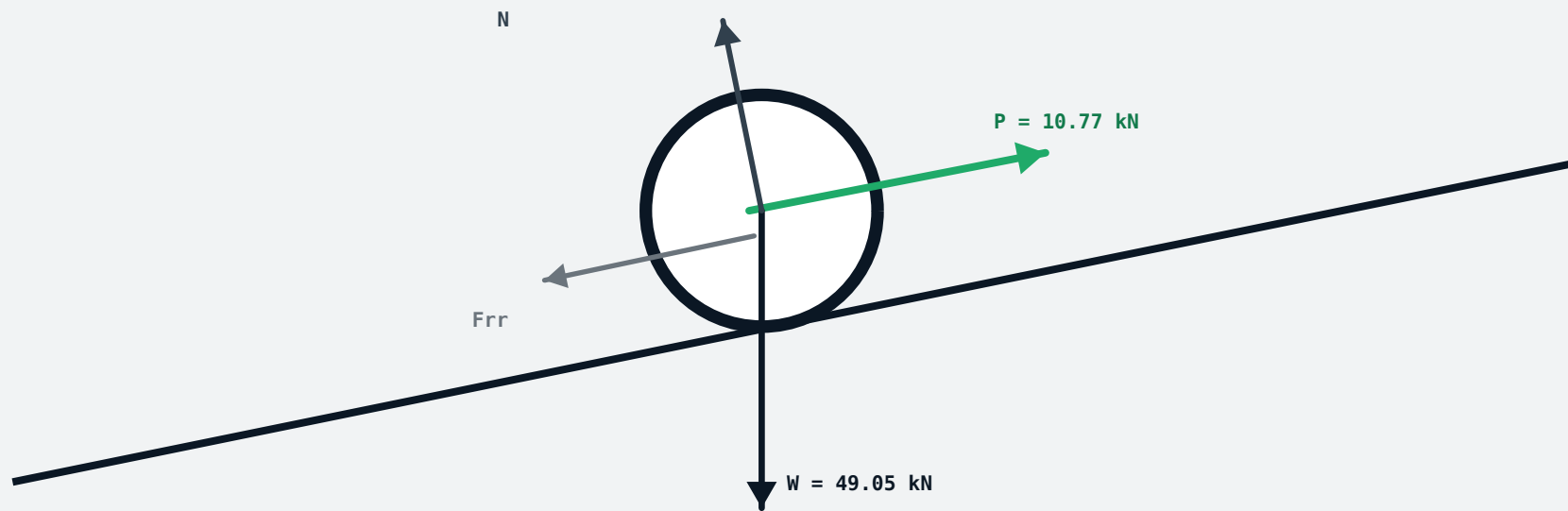


TECHNICAL READING

The energy needed to raise five tonnes by 100 m is modest. Energy alone does not explain why the task is impossible.

STEP TWO

Force, not energy, is impossible



QUASI-STATIC FORCE BALANCE · PUSH PARALLEL TO SLOPE · 11.54 deg

AVERAGE PUSH FORCE

10.77 kN

Approximately 1.10 tonne-force

MECHANICAL WORK

1.496 kWh

Per complete ascent

AT 100 W

14.96 h

Continuous output benchmark

INPUT-ENERGY EQUIVALENT

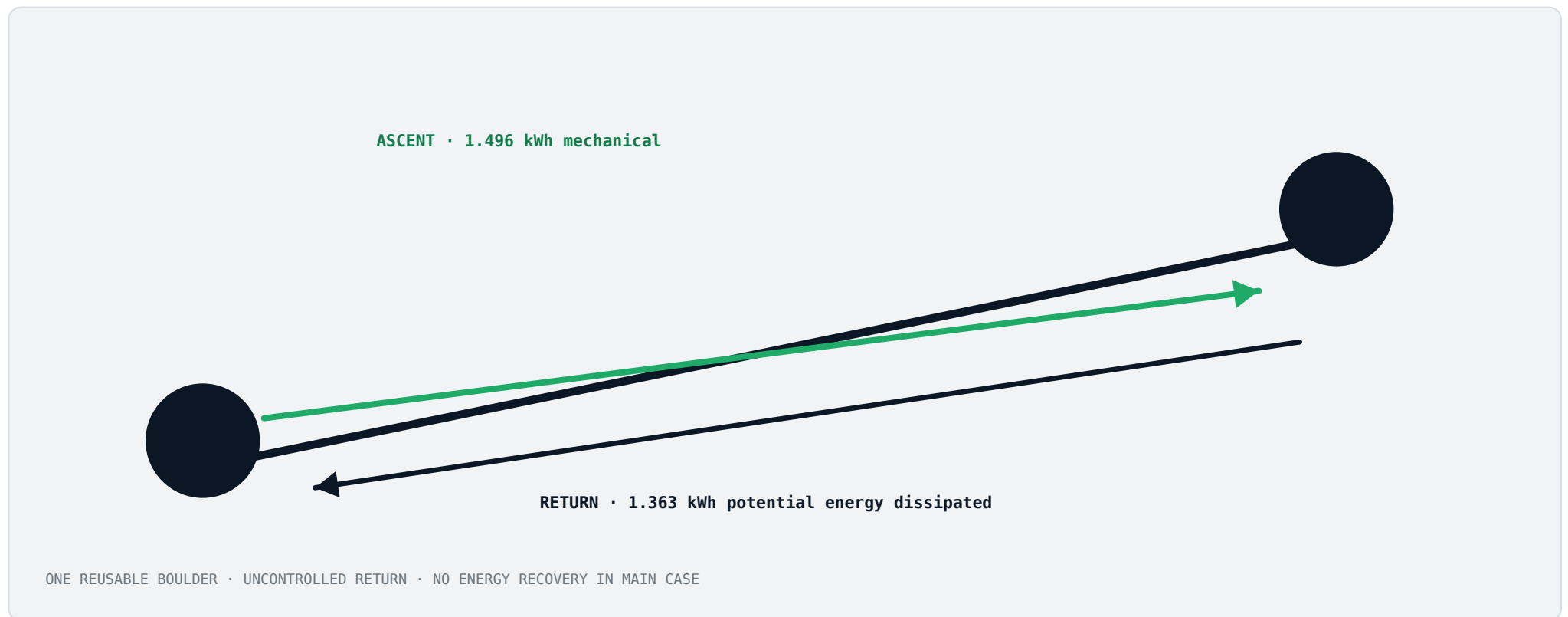
20% CONVERSION EFFICIENCY

$1.496 \text{ kWh} / 0.20 = 7.480 \text{ kWh}$ per cycle, equivalent to approximately 6,436 kcal of input energy.

The 20% efficiency is a screening calibration, not a physiological model of Sisyphus. The force requirement remains beyond an unaided human even though the energy order is familiar.

STEP TWO

The system resets itself



ONE CYCLE

500 m ascent

100 m vertical gain

1,000 CYCLES

500 km

100 km cumulative rise

RETURN LOSS

1.363 MWh

Cumulative potential energy

MATERIAL AND ENERGY LOOPS

MATERIAL LOOP

The same 5 t boulder returns to the start. No replacement is included over the reporting slice.

ENERGY LOOP

Potential energy is lost on descent. Each new ascent requires fresh input energy.

Reuse prevents the stone from scaling with cycle count. Operational input still scales linearly: repetition, not material throughput, controls the cumulative result.

STEP THREE

A deliberately narrow boundary

FLOW / COMPONENT	FACTOR	APPLICATION
Aggregates, primary material use	7.80307 kg C02e/t	One reusable boulder
Electricity generation	0.13096 kg C02e/kWh	Input-energy component
T&D	0.01299 kg C02e/kWh	Network component
WTT generation + T&D	0.04041 kg C02e/kWh	Upstream component
Combined consumption proxy	0.18436 kg C02e/kWh	Calculated total

EXCLUDED

INCLUDED

- One 5 t aggregate boulder
 - Potential-energy work
 - Rolling-resistance work
 - 20% input-energy conversion
- Food and physiology
 - Hill and track construction
 - Maintenance, dust and transport
 - Supervision and end of life

REPRESENTATIVENESS

DEFRA 2026 is strong for factor transcription and weak for Hades. Electricity is an accounting proxy, not a claim about the punishment's actual energy source.

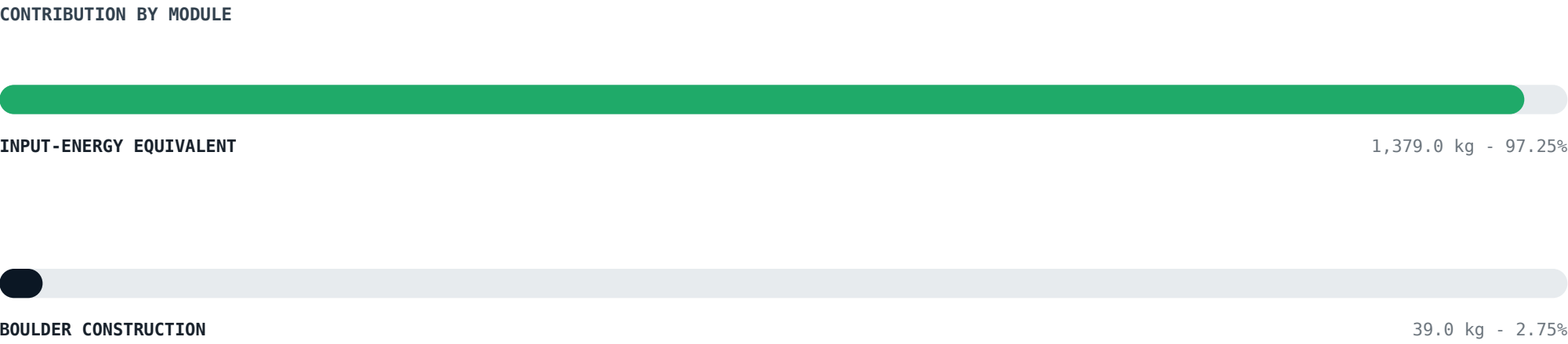
STEP FOUR

1.418 t CO2e per 1,000 cycles

MAIN RESULT

1.418 t CO2e

1.418 kg per cycle - 2.836 kg per slope-km



HOTSPOT STATEMENT

The boulder is reused. The repetition is not.

Over 1,000 cycles the energy proxy dominates, but the absolute result remains moderate because the mechanical work per ascent is small and no infrastructure is constructed.

STEP FIVE

Eternity is not a functional unit

CUMULATIVE EQUATION

GWP(N) = 39.015 kg + 1.379010 kg x N

CYCLES N	BOULDER	ENERGY	TOTAL
1	39.015 kg	1.379 kg	40.394 kg C02e
100	39.015 kg	137.901 kg	176.916 kg C02e
1,000	39.015 kg	1,379.010 kg	1.418 t C02e
10,000	39.015 kg	13,790.102 kg	13.829 t C02e
N -> infinity	fixed	unbounded	unbounded

INTERPRETATION

The stone approaches zero impact per cycle as it is reused. The energy term never does. A finite reporting slice is therefore a methodological necessity, not a convenience.

STEP FIVE

What changes the loop

RETURN CASE	RECOVERED PE	TOTAL	VS MAIN
Uncontrolled	0%	1.418 t CO2e	baseline
Partial capture	50%	0.790 t CO2e	-44.3%
Regenerative	70%	0.539 t CO2e	-62.0%
Ideal capture	100%	0.162 t CO2e	-88.6%

GEOMETRY RANGES

VERTICAL RISE: 50 -> 150 M

0.792 -> 2.043 t

Same 5 t boulder and 500 m slope

BOULDER MASS: 1 -> 10 T

0.284 -> 2.836 t

Same route and efficiency

SENSITIVITY LOGIC

- Mass scales construction, force and energy almost linearly.
- Vertical rise changes both potential energy and required slope force.
- Capturing the downhill potential energy is the strongest modeled design lever.

STEP SIX

Engineer the repetition

FINDING 01

One ascent requires only 1.496 kWh of mechanical work, but an average parallel push of 10.77 kN.

FINDING 02

Across 1,000 cycles, the input-energy equivalent contributes 1.379 t CO₂e and 97.25% of the total.

FINDING 03

With no final cycle, cumulative GWP has no finite endpoint. The result must always be tied to a declared reporting slice.

DESIGN PRIORITY

- Recover energy during the return before optimizing the stone's material footprint.
- Reduce boulder mass or vertical rise to lower force and energy together.
- Keep physiological and dietary impacts separate unless a defensible food scenario is selected.

Sisyphus is a repetition system disguised as a punishment.

LCA of the impossible - Season II

MYTHS & LEGENDS

**Where ancient stories meet
measurable emissions**